

# AHMED BOUBAKRY

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## SUMMARY

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Robotics and computer-vision engineer (M.Sc., EPFL) researching perception systems for embodied agents at the MIT Camera Culture Lab. Hands-on across detection, segmentation, and sensor fusion for real-world autonomy, with PyTorch, C++, and ROS2. Particularly interested in computer-vision and image-processing applications, from drone imagery to rover navigation.

## EXPERIENCE & RESEARCH

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**Visiting Student** 2026 · until Sept.  
MIT Camera Culture Lab Cambridge, MA

- Master's thesis project: built AutoCI, an agentic system that designs task-specific imaging systems end to end, jointly choosing the sensing modality, optics, and perception algorithm through a propose-and-validate loop between two LLM agents reasoning from first-principles physics.
- Cut search cost by executing each proposed pipeline instead of retraining it per candidate, removing the inner training loop that dominates budget-limited co-design.
- On two harder co-design tasks, the generated designs reached 95.6% accuracy on robotic grasping and 90.8% on object localization in cluttered, occluded scenes. First-author paper under review at AAAI-27.

**Computer Vision Engineer Intern** 2025 · 6 mos  
Michelin Recherche et Technique France

- Developed computer vision for a robot that autonomously navigates to vehicles and inspects their wheels: wheel detection, defect detection, and vehicle weight estimation from onboard sensing.

**Computer Vision Semester Project** 2024 – 2025 · 4 mos  
LUTS, EPFL Urban Transport Systems Lab Lausanne, CH

- Segmented roads and occlusions from drone video with Segment Anything 2 (SAM2), reaching 83.2% detection accuracy in obstructed scenes.
- Built detection and analysis of irregular vehicle components for an autonomous inspection system, using depth cameras and onboard sensors; graded 6/6.

**Stroke Lesion Segmentation Semester Project** 2025 – 2026 · 4 mos  
LTS5, EPFL Signal Processing Laboratory 5 Lausanne, CH

- Researched synthetic stroke-lesion simulation and deep-learning segmentation on Human Connectome Project (HCP) MRI data.
- Raised segmentation accuracy by ~10% by augmenting scarce annotations with simulated lesions; graded 5.75/6.

## EXTRACURRICULAR

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**Rover Navigation** 2023 – 2024 · 8 mos  
EPFL Xplore Lausanne, CH

- Implemented LiDAR–depth–camera sensor fusion, extending detection range and improving localization accuracy.
- Optimized sensor integration for the European Rover Challenge, contributing to a top-10 finish.

**Teaching Assistant, EPFL.** TA for Linear Algebra, Analysis I and II, and Advanced Analysis; evening tutor for the Centre Propédeutique. 2021 – 2025

**MOOC Editor, EPFL.** Proofread and refined the exercises and solutions for the Analysis 1 online course. 2023 · 1 mo

**Class Representative, EPFL.** Section delegate representing peers across the academic year. 2022 – 2023

**Expedition Organizer, GenoRobotics.** Secured sponsors and funding for the 2024 Madagascar expedition field-testing DNA-extraction prototypes. 2023 – 2024 · 1 yr

**Events Manager, Dynamic EPFL.** Ran fundraising events (bake sales, raffles, crowdfunding) for the study trip. 2022 – 2023 · 1 yr